

ChatGMP: A case of AI chatbots in chemical engineering education towards the automation of repetitive tasks

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ABSTRACT

Artificial Intelligence (AI) is rapidly and consistently becoming more integrated in various aspects of our lives. One of the areas where these systems are increasingly used is education. In fact, it is both being incorporated into specific curricula, allowing students the possibility to acquire skills within this field, and more recently AI has been used as a tool to facilitate the teaching and learning process. However, an increased demand and availability of these tools do not imply a successful switch from traditional to AI-supported learning. In this work, ChatGMP, a chatbot leveraging a Large Language Model (LLM) able to conduct an interview exercise in a Master's Degree course taught at the Technical University of Denmark (DTU), is introduced. The exercise consists in a student interview of a fictitious company, represented by the teachers or ChatGMP, regarding its Good Manufacturing Practices (GMP). The aim is for the students to ask sensible and well-reasoned questions to acquire the necessary documentation to make an exhaustive report indicating whether the company is a potential fit for business. To evaluate the initiative, we compare the performance of ChatGMP to the one of the physical teachers of the course, as well as the perception of the students towards it. The results show no significant difference in the information provided by the teachers and the model, enabling the students to achieve similar learning. The students that interacted with ChatGMP are satisfied with the initiative and would likely recommend future students to perform the audit with the digital tool. This initial experiment and its positive results lay the foundation for opening the discussion on how to use LLMs in education, the opportunities they could provide, as well as their limitations and drawbacks.

1. Introduction

Education 4.0 is at the intersection of digitalization and Industry 4.0 and represents a transformative shift in learning and teaching paradigms. Conventional educational models are no longer deemed enough; they present several challenges, such as a one-size-fits-all approach that disregards differences in pace, learning style, interests, and needs, alongside a lack of adaptability, particularly to technological change (Labadze et al., 2023; Stöhr et al., 2024; Rodrigues et al., 2024; Alfredo et al., 2024). This often leads to the compounded effects of disengagement and poor learning outcomes. Thus, to pivot from traditional education, we need to update the curricula to reflect current needs and integrate cutting-edge technologies like generative Artificial Intelligence (AI) as seamlessly as possible in classrooms. Recent years have

witnessed the rise of Large Language Models (LLMs), which are natural language processing (NLP) models trained on massive amounts of text, developing a broad general knowledge that can be applied to different fields and applications (Brown et al., 2020). LLMs such as ChatGPT (Brown et al., 2020; OpenAI, 2024) and Microsoft Copilot (Microsoft, 2024) are progressively becoming an integral part of our lives (Stöhr et al., 2024). People use them daily to ask general questions, write entire paragraphs, as well as, generating computer code (Schweidtmann, 2024). Even companies are incorporating paid versions or deploying bespoke models to answer their specific needs; chatBASF (BASF, 2023) and chatNN (NovoNordisk, 2024) are examples of chatbots created by companies for their in-house use. In education, teachers are now exploring the possibility of automating repetitive and laborious tasks through AI to reduce the teachers' workload and provide customized evaluation

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Table 1
AI-driven chatbots most commonly used in education.

Chatbot	Launch year	Main application & goal	Refs.
Piazza	2009	Aid collaboration and discussion in academic institutions regarding assignments and course content	(Ruthotto et al., 2020; Wang et al., 2020)
Socratic	2013	Create a community to make learning accessible to all students. Designed to help students learn new concepts. It has been acquired by Google in 2018.	(Alsanousi et al., 2023; Moppel, 2018; St-Hilaire et al., 2022)
Habitica	2013	Help students develop and implement good studying habits by gamifying the learning process. It converts to-do list into game-like experiences.	(De Sales & Antunes, 2021; Zhang, 2023)
Replica	2017	Designed to be a social-based chatbot, acting as companion and friend to students; it can listen to problems and offer advice.	(Pentina et al., 2023)
Ada	2017	Provides customized tutoring to students, by answering questions and providing personalized feedback.	(Konecki et al., 2023)
ChatGPT	2022	Created by OpenAI. Chatbot that can generate text, create diverse creative content and provide information in the form of answers to queries.	(Dergaa et al., 2023; Khademi, 2023; Rudolph et al., 2023)
Bard	2022	Created by Google. Chatbot that can generate text, language translation, produce creative content, and provide information in the form of answers to queries.	(Rudolph et al., 2023)
Copilot	2023	Created by Microsoft (previously named Bing Chat). It is integrated into Microsoft 365 in an effort to aid educators and students by improving learning experiences and productivity.	(Rudolph et al., 2023; Moradi Dakhel et al., 2023)

(Rodrigues et al., 2024). Furthermore, using generative AI seems to be a promising way to tailor the classes and teaching materials to different student profiles and learning needs (Rodrigues et al., 2024; Divya et al., 2023; Wang & Demszyk, 2023).

In this work, we aim to investigate the usage of field specific LLMs as part of the curriculum in higher education which ultimately targets to: (i) incorporate active teaching elements in large classrooms (> 100 students); (ii) substantially reduce time spent by the teachers on repetitive tasks; (iii) provide an interactive and fun learning experience for the students; and, (iv) assess the capacity of trained LLMs to answer high-level open-ended questions. Thus, the manuscript presents *ChatGMP*, a chatbot based on a pre-trained LLM and introduced into the curriculum of a Master's Degree course in Good Manufacturing Practice (GMP) at the Technical University of Denmark (DTU). *ChatGMP* is able to perform an audit exercise where students ask GMP questions and the chatbot answers them coherently and supplies the necessary documentation to write a report of the findings. The chatbot is provided to the students through an interactive Graphical User Interface (GUI). To evaluate the initiative, the performances of *ChatGMP* and the physical teachers are compared through a series of quantitative and qualitative analyses.

The manuscript is structured as follows: in Section 2 a short literature review on the usage of generative chatbots in education is presented. Thereafter the design and underlying methods of the implemented *ChatGMP* are introduced in Section 3. In Section 4, the results of the experiment are presented, including students' and teachers' perspective about the experience. We discuss the learnings of the experiment in Section 5, highlighting the strengths and weaknesses, as well as possible future perspectives (Section 6) and limitations (Section 7). Finally, we provide an overall conclusion in Section 8 and the code in Section 9.

2. Background and aim

The focus of Section 2 is restricted to only include research in generative chatbots used in education and contributions in the literature published from 2019 onwards, using pre-trained generative models. The Section is divided into two parts: subsection 2.1 summarizes the trends and new developments in generative chatbots in education, subsection 2.2 introduces the course selected to test generative chatbots at our university.

2.1. Generative chatbots in education

Pedagogical theories indicate that differentiated and individualized learning is a highly effective teaching method, spurring educators to prioritize this approach across various domains (Landrum & McDuffie, 2010; Hwang, 2014). Continuous and immediate feedback has also

been identified as one of the most effective learning methods (Laurillard, 2009; Hattie & Timperley, 2007), helping prevent small errors and misunderstandings from escalating and settling. At the same time, the conventional education structure has been facing several challenges, which have been more evident in recent years, such as different students' learning paces and styles, the absence of customized attention to the students, the ongoing fight to keep up with the rapid evolution and transformation of technology, packed classrooms, and time-consuming administrative tasks (i.e., grading, scheduling, etc.) (Yan et al., 2024; Labadze et al., 2023; Caccavale et al., 2024a, 2024b; Chiu, 2024). Indeed, to solve some of the said challenges, AI-driven chatbots have been seen as a promising tool increasingly being adopted by educational establishments (Labadze et al., 2023; Stöhr et al., 2024; Rodrigues et al., 2024; Alfredo et al., 2024). However, it is worth noting that there is no overall consensus on this matter; therefore, there is a significant interest in this topic and in mapping and evaluating the advantages vs. disadvantages of using chatbots in education. This interest is accompanied by a growing body of literature devoted to analyzing the use of chatbots and the part they play in education, and this is likely to expand quickly due to the fast-paced evolution of generative AI. Some of the most employed AI-powered chatbots currently in education are summarized in Table 1. In contrast to most of the chatbots reviewed in Table 1, *ChatGMP* developed in this work shows deep domain-specific knowledge, which is tailored to the course it is applied to.

Since it is beyond the scope of this paper to give a comprehensive review of the use of chatbots in education (weighing threats against benefits), the reader is referred to recent original articles on the topic, such as Stöhr et al. (2024), Rodrigues et al. (2024), Espartinez (2024), Schulze Balhorn et al. (2024); and literature reviews such as Gao et al. (2024), Labadze et al. (2023), Deng and Yu (2023), Crompton and Burke (2023), and Kuhail et al. (2023).

2.2. Background on the selected course

DTU offers a course in Good Manufacturing Practice (GMP), the governing guidelines from the authorities of food and pharmaceutical safety that set the standards for production industries. According to the World Health Organization (WHO), GMP prevents errors that cannot be eliminated through quality control of the finished product, and without which, it would be impossible to ensure that every unit of a medicine is of the same quality as the units of medicine tested in the laboratory.¹ To ensure that companies are working according to GMP guidelines and

¹ Medicines: Good manufacturing practices, available at: <https://www.who.int/news-room/questions-and-answers/item/medicines-good-manufacturing-processes>, Accessed: 29-07-2024.

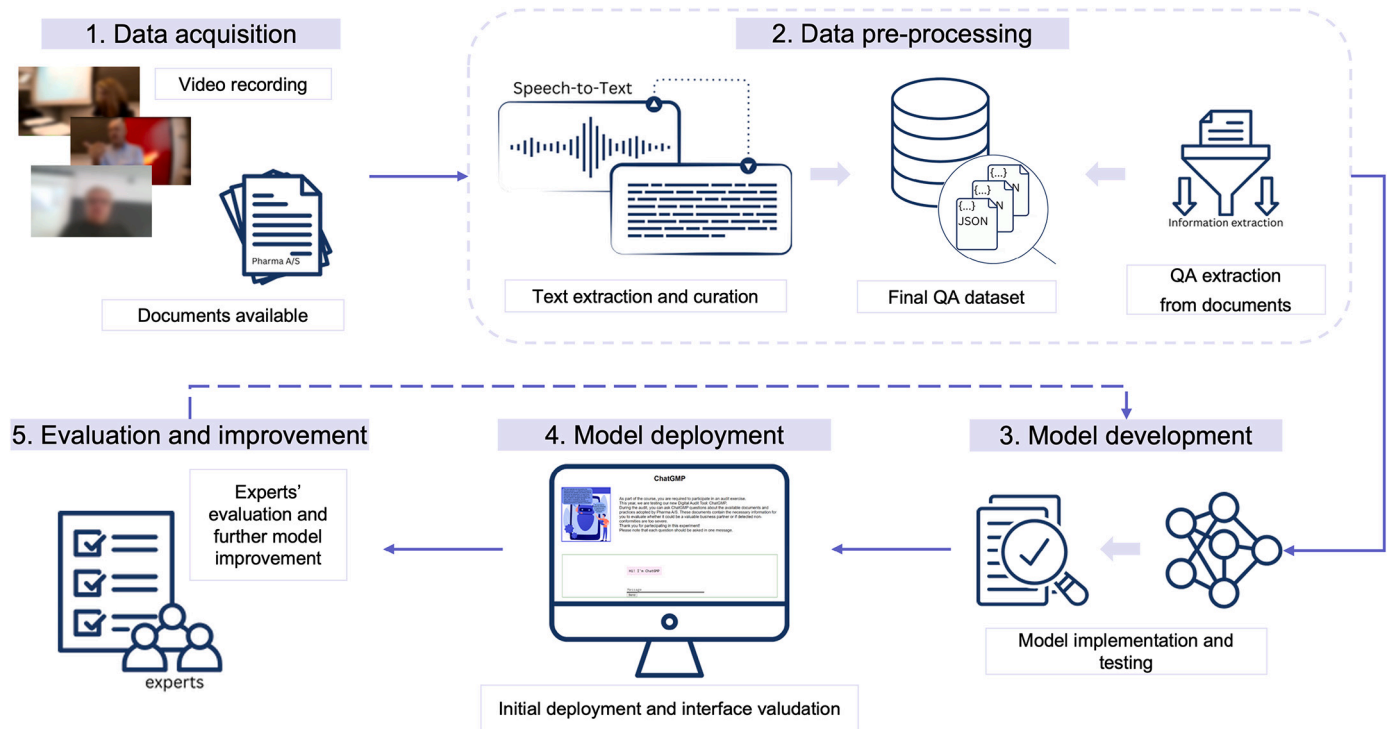


Fig. 1. Full model pipeline, from data acquisition to experts' evaluation.

implement required quality management systems, they are audited by authorities. Therefore, in pharmaceutical production, it is very important that these processes are regulated and students are well trained and educated on these regulations. As an active learning part of the course, students are required to plan and execute an audit at “Pharma A/S”, a fictional company created for the purpose of the exercise. In this exercise, they assume the role of an auditor and audit the company, which is represented by teachers. During the audit, students need to ask questions about the practices adopted by the company and request relevant documentation. This information is a prerequisite to evaluate whether the company in question could be a viable business partner or if detected non-conformities are too severe. The role of teachers is to provide accurate answers, in case the documents are available, or vague responses, leaving students to reflect on the company’s behavior. The hand-in for this exercise, which is not the final exam but just a group assignment that needs to be passed in order to be allowed to the exam, is an audit report. The report is evaluated to assess the groups’ abilities to detect minor and major non-conformities as well as aligning them to the respective legislations. In a final statement, they need to reveal if the company is compliant, and if not, a necessary action plan needs to be provided.

The course is administered once per year in the spring semester and around 120 students are allowed to register. However, usually at least 25% more students express an interest in enrolling in the course. The only limitation to the yearly uptake is the audit exercise, since the three teachers need to personally perform the interviews, and every year around 21-24 groups of 5-6 students each are formed. Having administered this exercise for various years, the teachers claim that performing 7-8 audits each can become repetitive and energy draining, since the questions and the topics discussed are overall quite similar. Therefore, automating this process would benefit both students, increasing the yearly uptake and thus giving the possibility to more students to enroll, and teachers, automating a time-consuming and repetitive task. The course is generally discursive, and the regulations and documentation are straightforward to discuss, so it poses a suitable use case for the application of AI. Moreover, different studies, such as (Caccavale et al.,

2024a), show that students are open and willing to test AI in these types of applications.

This work presents the first trial of substituting the role of the physical teacher with *ChatGMP*, an AI-powered digital audit tool to represent the fictional company. Therefore, the objective of this study is to develop *ChatGMP* and compare the effectiveness of the chatbot with the performance of a live teacher in answering audit questions as “Pharma A/S” (the fictitious company created for this specific exercise). Among the further added benefits that introducing an LLM to the course could generate, is that students might be more engaged by the gamified nature of the interaction and might be more inclined to ask for further clarification of the given answers. To be noted is that this initiative does not aim to move towards a teacher-less course, but rather to help teachers focus on the lectures and material preparation instead of spending time on exercises that are very time-consuming and repetitive. It is also important to take into consideration that individual students may have different preferences, whether they are comfortable interacting with a chatbot or favor a physical teacher, and therefore, at least for the first few editions of the course, the participation in this experiment will be fully voluntary, so that groups have the choice to perform the audit with a teacher or with *ChatGMP*.

3. ChatGMP: data and model development

This section introduces the technical aspects behind *ChatGMP*, from data acquisition to the evaluation and further improvements of the deployed model. Some necessary terminology is introduced in subsection 3.1. It follows an explanation of the data acquisition and pre-processing in subsection 3.2, the model development details in 3.3, the model deployment in 3.4 and the final evaluation pre-experiment in 3.5. The full model pipeline is shown in Fig. 1. *ChatGMP* presents a *back-end* and a *front-end* component. To facilitate the understanding of these components, and to help researchers better grasp which of these were implemented in this work and which can be readily found online, the main aspects are summarized below.

- **Back-end:**

- **Curated, domain specific dataset:** The data is stored in a dataset in JSON format, a `<QUESTION>` is the key and its `<CONTEXT>` is the value. The dataset was collected, pre-processed and curated by the authors.
- **Pre-trained model:** A pre-trained, open-source LLM, FLAN T5 (Chung et al., 2022) was used. The decision on using this model is based on benchmarking other selected, widely available, pre-trained models. These models are open-source and the authors did not change the model architecture.
- **Prompt engineering:** The pre-trained LLM is prompt engineered; the prompt given to the model is composed of a `<QUESTION>` and enriched by the `<CONTEXT>` containing the historical answer. The `<CONTEXT>` is found through semantic search by calculating the cosine similarity between the query question (asked by the student) and the reference questions (historical questions stored in the dataset). This part of the code was implemented by the authors.
- **Deployment:** The code is deployed locally using the Flask package in Python. The code to deploy the webpage hosting *ChatGMP* was written by the authors following the Flask package - documentation is very easy to follow and replicate.
- **Front-end:**
- **Interface:** The interface was created in HTML by the authors.

3.1. LLM terminology

LLMs are able to take natural language (or human written instructions) as input, which is known as *prompt*. The *context window* available to the prompt is generally quite large; therefore, the instructions given to the LLM can be elaborate. In this work, in-context learning is performed, since the prompt is enriched with some context to improve the quality of the generation. Practically, the model performs a next-word prediction task. The output generated by the model is called *completion*, while *inference* refers to the act of using the model to generate text; in this case it is zero-shot inference, since the LLM is able to perform well on the task without needing further examples.

3.2. Data acquisition and pre-processing

In order to train the model, domain specific data needs to be collected and curated. The data acquisition and pre-processing pipeline includes the following steps:

1. **Video recordings:** the data used in this work is mainly collected from the audit exercises; the audits were performed and video recorded in the spring semesters of 2022 and 2023, summing up to 35 audits of approximately 45 minutes length each.
2. **Audio extraction:** the audios were then extracted from the videos using the Python library *MoviePy*.²
3. **Diarization:** to detect the different speakers and therefore being able to identify the questions (asked by the students) and the answers (formulated by the teacher), diarization, a process of automatically partitioning an audio recording into segments that correspond to different speakers, was performed using *Speaker Diarization 3.1*.³
4. **Speech-to-Text:** the speaker segmentation was then used to extract the text through a speech-to-text model, OpenAI's pre-trained Whisper (Radford et al., 2023).
5. **Merge QA pairs:** once the text had been extracted and the different speakers had been detected, the teacher, which is usually the speaker with the most dialogue, was classified and all the students'

questions were merged, therefore resulting in a document with the question-answer pairs.

6. **Quality evaluation:** to ensure the highest quality of data to be used as model input, the resulting text documents (one for each audit performed), were manually checked and edited when necessary.
7. **Final QA dataset:** once the documents were curated, the question-answer pairs were concatenated and saved in a JSON file, for easiness of use.

The data-preprocessing pipeline is shown in Fig. 2. All steps to reach a final dataset of question-answer pairs are automated, except the quality evaluation step, where manual quality check was performed to ensure the highest quality of data possible.

In addition to the question-answer pairs collected from the audit exercises, the dataset was enriched with information extracted from the documents that students need to request and collect during the audit. These documents refer to general procedures in GMP or are visual examples of a template filled in by the employees of "Pharma A/S". The total number of documents available to students is eleven. These documents include standard procedures such as maintenance, logistics, training of operators, as well as an example of a calibration record and a cleaning template.

3.2.1. Data anonymization process

Students were asked to give their consent for the data collection and were made aware that the interaction would be anonymized and transformed into a text-based dataset with question-answer pairs, which will then be used in research. They were informed that their personal data would not be distributed, and that no further personal information was collected, therefore it would not be possible to trace the question back to any students. All data, including the video recordings, is safely and ethically handled according to GDPR.

The audits were originally recorded by video for two main reasons, (i) because students often ask for a recording of the audit, since they have to write a report with their findings and to ensure that they did not miss useful details, and (ii) to capture the whole interaction between the auditor (students) and the auditee (teacher). This is motivated by the fact that the final version of the tool will be an avatar or human-like agent, and therefore, the teachers' responses are necessary to train the model. Therefore, it would be impractical to only record the teachers' answers. Even in future versions of the model, the original students' recordings will not be used, to protect students' privacy. Finally, once the text is extracted from the audios, the resulting corpus of audits is extensively manually curated, both to ensure high-quality output, but also to prevent the presence of any defining attributes that can link to specific individuals.

3.3. Model development

ChatGMP is a question-answering (QA) LLM based on a pre-trained model and delivered to the students through an interactive user interface. The LLM component is based on a pre-trained, open-source model, which can be downloaded from the HuggingFace library⁴. Multiple Transformer-based (Vaswani et al., 2017), pre-trained, open-source⁵ models, both encoder-decoders (or sequence-to-sequence) and autoregressive (or decoder only) QA models, were benchmarked in order to find the most suitable model for the course. The performances of various models, including: T5 (base, large) (Raffel et al., 2020), FLAN-T5 (base, large, xl) (Chung et al., 2022), LLaMA 2 (Touvron et al., 2023) were compared. The final model selected to be used in the course was

⁴ https://huggingface.co/docs/transformers/en/model_doc/flan-t5.

⁵ Although LLaMA 2 (Touvron et al., 2023) cannot be strictly classified as open-source, the weights of the model can be downloaded and therefore the model can be used similarly to others.

² <https://pypi.org/project/moviepy/>.

³ <https://huggingface.co/pyannote/speaker-diarization-3.1>.

Data pre-processing

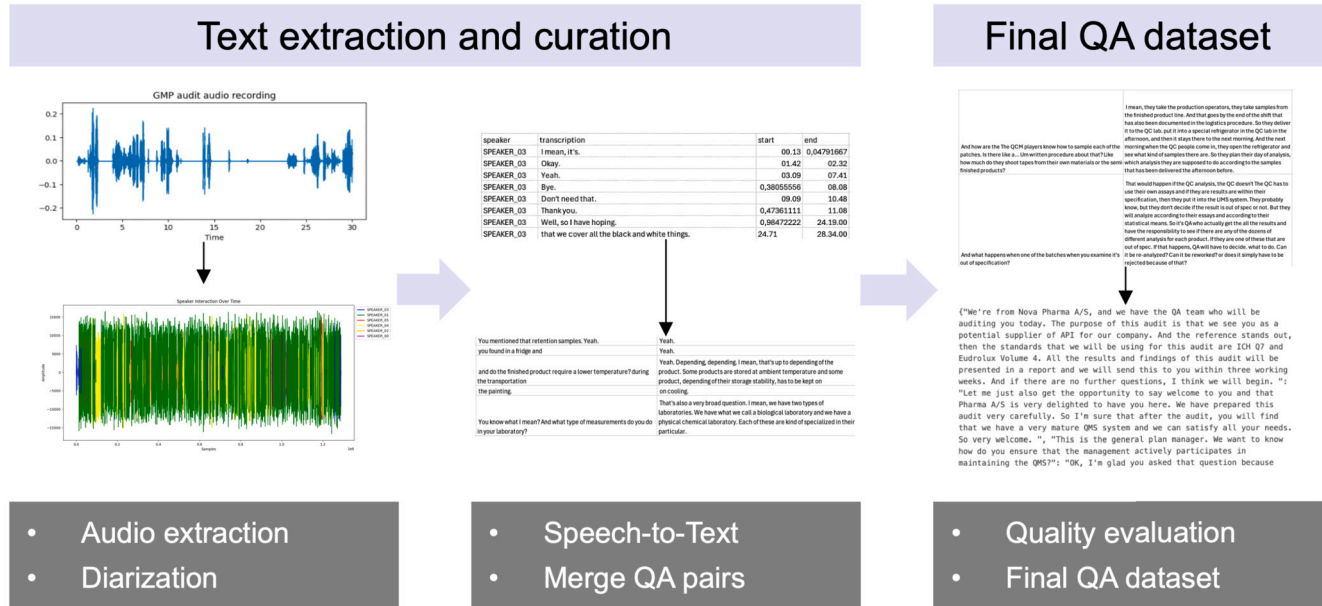


Fig. 2. Data pre-processing pipeline. The points in the boxes correspond to the specific pre-processing step described in Section 3.2. In the video captures, the teachers are taking part in the audit exercise and, therefore, responding to students' questions regarding GMP. The video captures have been blurred to protect the teachers' privacy.

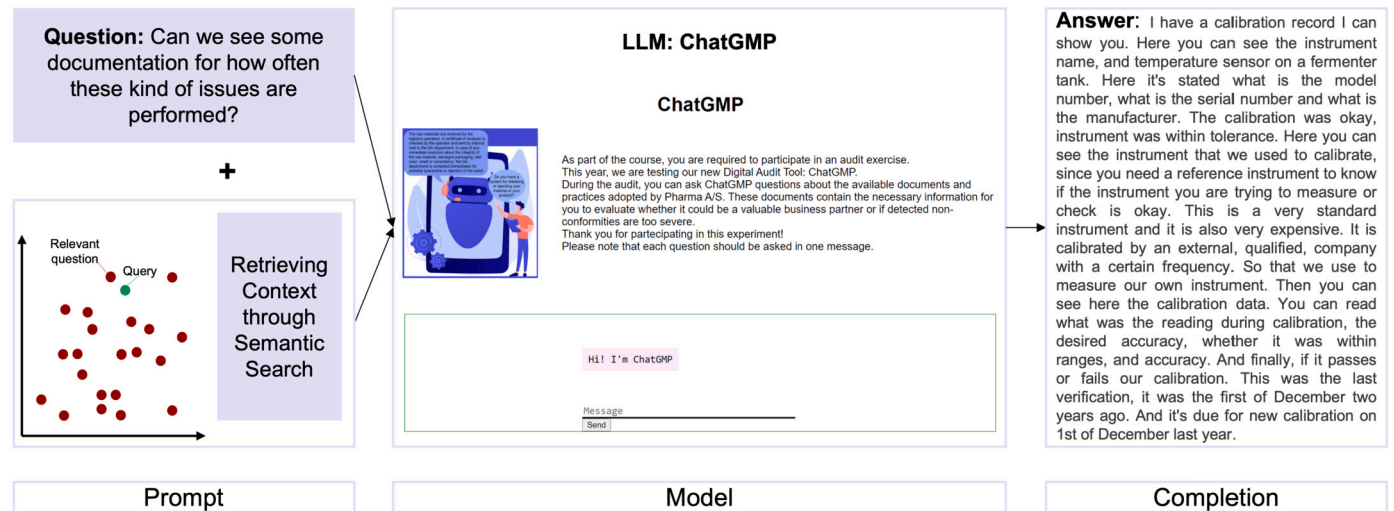


Fig. 3. Schematics of the prompt given to ChatGMP. In the given example, ChatGMP is prompted to open the "Calibration record" document (opened in another window as in Fig. 4), and summarizes it to the students.

FLAN-T5 base. The decision was a trade-off between performance and runtime efficiency: in fact, the model had to be run locally using only one GPU, therefore a model such as LLaMA 2, although more accurate in the generation, was too slow to be used with the students, given the available resources and infrastructure.

In our case, the prompt is composed of a question (asked by the student group performing the audit) and enriched with the context that contains the response (historical answer) to the question. The context is retrieved through semantic search by calculating the cosine similarity between the input question and the most similar question in the corpus. Once the most similar archival question is found, its answer, extracted from the historical audit exercises, is then appended to the prompt. The LLM then generates an answer to the question asked based on the context provided. The process pipeline is visualized in Fig. 3.

The decision to perform prompt engineering and retrieve the context through semantic search stems from our aim to restrict the probability of hallucination of the LLMs. In fact, the quality and exactness of the response that the model provides the students is of the utmost importance for the success of the audit exercise. In our case, prompt engineering has a double benefit: (i) it reduces hallucinations and therefore provides more relevant responses and adheres to the context provided; and (ii) improves the previously given answers: the dataset is collected from oral language, meaning that, in spite of the quality evaluation performed, it naturally contains small repetitions and mistakes which are typical of verbal utterances. This means that the model is able to generate answers that are syntactically more correct, and therefore no further manual curation of the answers is needed. Moreover, not fine-tuning the model also allows for saving computing resources.

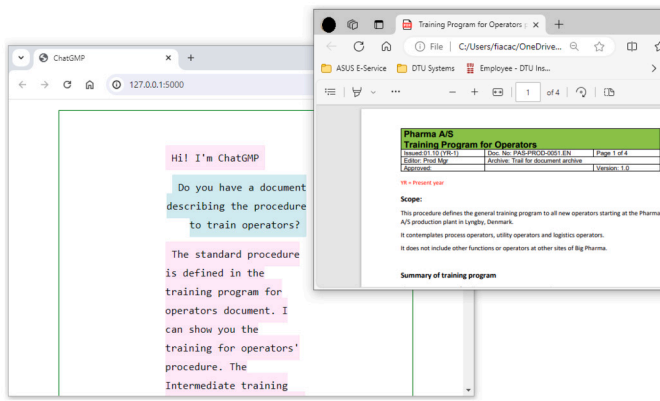


Fig. 4. Example of question asked by the students during the audit exercise with pop-up document shown.

Table 2

Average results of FLAN-T5 and LLaMA 2 models given the prompt (student's question) and context (teacher's answer). The results are calculated over a subset of questions (N=10) and A100 GPU was used.

	Cosine similarity	BLEU	Runtime
FLAN-T5 base	0.68	0.39	19.0 s
LLaMA 2	0.72	0.33	38.9 s

3.4. Model deployment

ChatGMP's interface, shown in Fig. 3 (step 2: "Model"), was developed in Python, using the Flask package for deployment. The front-end is in HTML. The model was deployed locally using a single NVIDIA GeForce RTX 2060 with Max-Q Design GPU. During the audit exercise, students interacted with the chatbot through a laptop connected to a monitor. One student was asked to type the questions and all students in the group could see the response on the monitor. If the students requested to see a specific document, the model was triggered to show the document through a pop-up window. Students then had the opportunity to skim the document before proceeding to the next question. An example of this is provided in Fig. 4. All the documents shown to the students during the audit were later sent to them.

3.5. Evaluation and improvement

3.5.1. Model performance

To evaluate the responses generated by the models, we calculated two metrics widely used for QA models, the BLEU score (Papineni et al., 2002) and cosine similarity. BLEU is a precision-based metric originally developed for evaluating machine translation and it quantifies alignment between the generated and the reference text. Cosine measures, such as similarity and distance, are commonly used in NLP to calculate the relationship between words, sentences or documents, as in Mikolov et al. (2013). The results of two selected models among the benchmarked ones, which best highlight the trade-off between accuracy and efficiency, are shown in Table 2. The cosine similarity is quite high, meaning that the generated output is similar to the original answers. Table 2 also shows that the BLEU scores are not as high as the cosine similarity; however, this does not necessarily mean that the model is not able to perform the task. The metric compares the set of tokens produced to the ones in the original answer, and therefore possible explanations for the low BLEU scores could include that the model tries to exclude redundant information or generates synonyms of the words in the original answer. Moreover, the used metrics, especially the BLEU score, are affected by the length of the completion compared to the original answer. Table 2 also showcases a comparison of the runtime necessary for

one completion. It is important to notice that, generally, larger models have a longer runtime, meaning that they will output the answer more slowly. This is evident by comparing the runtime of the two models tested, where FLAN-T5 is significantly faster to generate the completion than LLaMA 2. This has to be taken into consideration and addressed properly to ensure a fast and smooth experience for students.

3.5.2. Pre-experiment evaluation

Evaluating a model qualitatively, as well as quantitatively (as done above through the BLEU score and cosine similarity), is necessary before deploying it as part of the educational system. Before the audit exercise, students were asked to hand-in a document in preparation for the audit, which also included the questions the group planned to ask. Normally, the teachers would look at the questions before the audit to ensure that they were prepared to answer all of them. However, not to bias the results, *ChatGMP* was evaluated with questions from groups enrolled in the 2024 edition of the course, but excluding the 3 groups that decided to perform the audit exercise with it. This means that at the time of the audit exercises, the set of questions prepared by the 3 groups were unseen to *ChatGMP*. The model was tested before being used in the course both with regular users (colleagues being familiar with GMP concepts) and with expert users (the teachers). The feedback provided, which mostly consisted of minor bugs such as documents not opening, was then incorporated in the deployed version of the model. Moreover, some topics the model lacked knowledge of were discovered, and further data was added to the database, so that the model could then answer those questions in the actual audit exercises.

3.5.3. Statistical tests

After the audits, to assess whether the difference between the two groups (students performing the audit with a teacher and students performing the audit with *ChatGMP*) is significant, two statistical tests were performed. The null hypothesis is that there is no significant difference between the two groups. This would mean that the two groups are equally satisfied with their respective experience and was able to fulfill its objectives. For the first three research questions (RQ1, RQ2 and RQ3) we performed a Mann-Whitney U test (also known as the Wilcoxon rank-sum test) (Mann & Whitney, 1947), which is a non-parametric test that compares the distributions of two independent groups. It is commonly used for ordinal data and does not require the assumption of normality. For the last research question (RQ4) a Chi-square test (McHugh, 2013), which assesses whether there is a significant association between two categorical variables (in this case, the group and the response category) was performed.

4. Results

4.1. Audit results: students' perspectives

In the 2024 edition of the GMP course, students were divided in 21 groups to perform the audit exercise. The groups were asked to volunteer to test *ChatGMP* and perform the audit with it. Out of the 21 groups, 3 volunteered to test the chatbot. The opinions of the students after the audit exercise are summarized in Fig. 5. The responses are from students that performed the audit with a teacher (14 students) and *ChatGMP* (13 + 7 students). The feedback towards *ChatGMP* was collected in two surveys: one administered on the same day of the 3 audit exercises performed with it, on 21/03/2024, and in the general survey targeting all students enrolled in the GMP course and administered on 02/05/2024. For transparency, both responses concerning *ChatGMP* are presented. Those responses might be overlapping.

The plots show that students interacting with *ChatGMP* rate their experience similarly to the control group (students performing the exercise with the teachers). Especially for the question RQ1, "How happy are you with your experience?", presented in Fig. 5 (a), students seem to overall agree across the two groups. Regarding the second question about

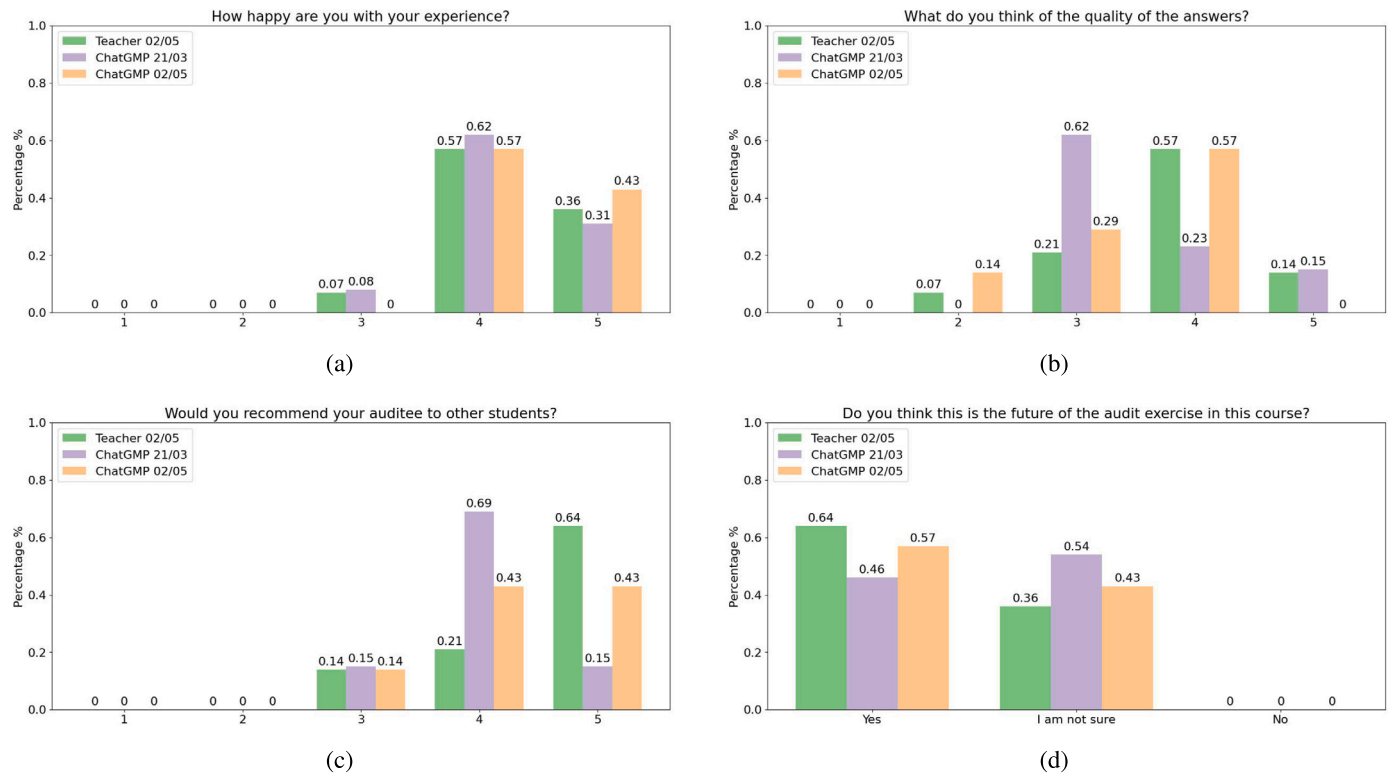


Fig. 5. Responses to surveys investigating the audit results both from students that performed the audit with a teacher (14 students) and *ChatGMP* (13 + 7 students). For transparency, we plot both responses concerning *ChatGMP* obtained right after the audit exercises in a survey given on 21/03/2024 (13 students, middle set of bars) and those that replied to the general survey administered on 02/05/2024 (7 students, last set of bars). Those responses might be overlapping. (a) Responses to the research question “How happy are you with your experience?” (RQ1). (b) Responses to the question “After having performed the audit, what do you think of the quality of the answers?” (RQ2). (c) Responses to the question “Would you recommend doing the audit with your auditee (teacher or *ChatGMP*) to other students?” (RQ3). (d) Responses to the question “Do you think this is the future of the audit exercise in this course?” (RQ4). RQ1, RQ2 and RQ3 are on a scale in range 1-5, from lowest to highest.

the quality of the answers, RQ2, the results in Fig. 5 (b) show that students in the control group mostly consider the quality of the answers to be good, with a few believing it to be very good, and a combined 28% to be average to low. For the *ChatGMP* sample answering the survey right after the audit exercise, students mostly believe the quality of the answers to be average, with a combined 38% of them to be good or very good. RQ3, the third research question shown in 5 (c), “Would you recommend your auditee (teacher or *ChatGMP*) to other students?” shows a bit more polarity. If the control group appears to be extremely convinced about their auditee (the teachers), the other is satisfied but does not show the same conviction. This is not necessarily a negative finding for the experiment, but might simply mean that some students need more time to adjust to changes and see the potential that AI could bring to education. Finally, the last question (RQ4) in 5 (d), “Do you think this is the future of the audit exercise in this course?”, shows substantial agreement across the different groups interviewed, suggesting that all groups were either satisfied with their auditee choice, or think that the other option might still be viable. This is also an interesting insight and it could be worth in the future to have the same groups perform the audits both with the teachers and *ChatGMP*, to be able to better compare their experiences. Interestingly, students answering the survey after a few weeks since the experience, have generally a higher perception of the audit exercise with the chatbot and of the quality of the answers. This can be seen across all the reported answers. We could hypothesize that, perhaps, the students realized that they were, in fact, given all the tools to produce a good report, and therefore developed a better opinion on the tool, which might have felt a bit frustrating at times, also given the stress of having to perform the audit and deliver a report. The average and standard deviations of RQ1, RQ2 and RQ3 are reported in

Table 3

Mean and standard deviation of the groups auditing the teacher and the two groups auditing *ChatGMP* for RQ1 (“How happy are you with your experience?”), RQ2 (“After having performed the audit, what do you think of the quality of the answers?”) and RQ3 (“Would you recommend your auditee to other students?”).

	RQ1	RQ2	RQ3
	avg. std.	avg. std.	avg. std.
Teacher	4.29; 0.59	3.79; 0.77	4.50; 0.73
ChatGMP1	4.23; 0.58	3.54; 0.75	4.00; 0.55
ChatGMP2	4.43; 0.49	3.43; 0.73	4.29; 0.70

Table 3. For RQ4, these metrics are not reported since the responses are categorical.

The distributions of the group performing the audit with the teacher to the other two groups, the students performing the audit with *ChatGMP* responding on 21/03 (D1) and the students performing the audit with *ChatGMP* responding on 02/05 (D2) are compared. The test statistics and p-values are reported in Table 4. For all questions, the p-value is more than the significance level (set to 0.05), so we fail to reject the null hypothesis and conclude that there is no significant difference in the distribution of responses between the two groups. Therefore, both Table 3 and Table 4 highlight that the responses of the two groups of students are comparable and show no statistically significant difference.

Table 5 presents more elaborate answers to the previously discussed questions, as well as possible advantages and limitations of the implemented chatbot.

Table 4

Mann-Whitney U test comparing the distributions of the groups auditing the teacher and the two groups auditing ChatGMP. Mann-Whitney U test is performed for RQ1, RQ2 and RQ3 and Chi-square for RQ4. The differences are not statistically significant.

	RQ1	RQ2	RQ3	RQ4
	U_M p-val	U_M p-val	U_M p-val	Chi2 p-val
D1	13.0; 1.00	13.5; .91	13.5; .91	0.3; .58
D2	16.0; .50	14.0; .83	14.0; .83	0.0; 1.0

Table 5

Further elaboration from the students of the course (3 groups). The first column presents the answers from students that performed the audit with a teacher (14 students), second and third columns represent answers with ChatGMP (13 + 7 students). For transparency, we present responses concerning ChatGMP obtained right after the audit on 21/03/2024 (13 students, second column) and those that replied to the general survey administered on 02/02/2024 (7 students, third column). Those responses might be overlapping. Redundant or not suitable answers are not presented.

Question	Teacher 02/05	ChatGMP 21/03	ChatGMP 02/05
What do you think of the quality of the answers generated by the virtual tutor?	- The answers of the auditee were not as we have expected. Some answers made no sense and felt like it was misdirecting us instead. - I believe that we received all of the answers in detail to our questions during the audit. - the teacher gave us great answers and documents related to our questions - Not 5, because I wasn't sure, whether the teacher really couldn't answer or whether the teacher wanted us to ask further into detail. This was cleared after a little confusion. - It was a good real experience to know how audit works. - Like the interaction. - The teacher answered as would I have expected from a real auditee. Of course, the answers were not always satisfactory to the auditors. But the teacher was very considerate and tried to give us hints about what we could ask to get the desired answers. - The answers were generally informative and clear with lack of details as they are meant to be noticed by us.	- Some answers were good, but sometimes we got the wrong answers which could be our fault as we didn't formulate more direct questions - way better than I expected. Extremely impressed - The answers seemed to be pre-determined so it didn't lie about stuff but it was sometimes a bit hard to get exactly what we want, maybe it could say more exactly when we are running in a direction it can't answer - it is very much real-style answers, even with the filler and the informal style answers so that is good. - it seems better than what i expected to be. We got the documents we needed. And it was very helpful. - I think the answers are detailed, but the documents the robot can provide are still kinda limited. And it seems some problems were a little misunderstood. - As it is an AI model, sometimes it did not understand our question, and we had to phrase it on a very specific way for it to give us the information we required.	- Sometimes it gave the same answers as in previous questions and many times, while asking questions regarding different topics, it provided documents already given while answering a previous question regarding a completely different topic than the question asked at the moment - The overall flow is good, but ChatGMP sometimes gives the same answer and the wait for an answer, while not too long - around thirty seconds or so - is an odd experience, as interacting with a real person usually gets you feedback immediately. - It was a new and interesting experience. I am excited to see that AI presence in bio courses. We had problems getting the same answers from ChatGMP for different questions. Some of the answers were contradictory to document scopes. Also typing and reading as the only way of communication wasn't comfortable. Adding audio communication in both ways would improve the experience a lot. - I think in some questions, we didn't get the exact answer we were expecting from ChatGMP.
Do you think this is the future of the audit exercise in this course?	- You can have more students and with the improvements in AI I think you develop a good program - If we are going to the industry, audit is a very important part, it is good for students to have a concept about how to apply GMP in the real cases. - What is the future? Doing the audit or doing it with ChatGMP? For the latter I would say no, because I really needed the small interaction with the teacher - I like the physical audit, but I did not try the chat one. I think that the chat one might be good if it means that the group sizes could be decreased. - Which form will be the future eventually depends on the audit objective. The main goal could be just to simulate asking questions and receiving the answers, which the ChatGMP could fulfill, however, the direct interaction with a person during an audit can never be simulated with this audit form. So depending on the accuracy of the audit simulation, either form might be preferred. - It's a theoretical course and there are not many teachers for this.	- I can see it happening, but I still think that real humans have more experience - I think it would work, but you need to make sure that they are asking short and precise questions. And got more time than just 45 min. - It is efficient if every group can do the audit with the chatGMP at the same time instead of a teacher. - I think it depends on the availability and perception of the auditor. - Because it works well, obviously could be a bit faster but I liked it - I still think you need somebody supervising, but it reduces workload to teachers - i think response is slightly slow in ChatGMP. But its good and compared to human I think its like same. - I think it can definitely be used as part of the course, but if the purpose is to carry out a "fake" audit, then it is better to interact with humans during it	- It is definitely an interesting approach, but ChatGMP should be further developed and optimized, so it could be implemented fully in the course - I very much enjoyed the whole experience of being the auditor with ChatGMP and working on the report after the audit. This activity familiarized me to many different concepts within GMP. I believe that using ChatGMP will be very helpful to the lectures to perform these audit exercises. - Even though it really facilitates procedures, I'm not sure if it could ever replace a company - It depends on which format the student actually prefers, and I think there are advantages to each. - Definitely would recommend it, it is the only course with the presence of AI for group exercise, fascinating and innovative. It helps to structure the train of thought. I think it is the future as at a certain point more groups will be able to conduct audits at the same time, and groups will get the same data. - Because, it depends on the people how they use AI tools.

Table 5 (continued)

Question	Teacher 02/05	ChatGMP 21/03	ChatGMP 02/05
Could you write some additional impressions?	- I feel like it was a great experience and put GMP into another perspective rather than only classes - It really felt like the auditee/teacher was part of a QA team of a company. - It was good - The teacher did a great job setting the scene and help us get into the role of the auditors with presenting all the people “present” in the audit. - The preparation was detailed and the time was well scheduled.	- Sometimes we needed to ask direct questions, whereas if it was a real [person] it would have understood us. - Overall a good experience. Bit stressful to do the typing! You should’ve shorten our questions. - The ChatGMP replies as a real person, but it doesn’t quite feel like a real conversation. - It was great experience to do audit with AI. - It was nice but a bit artificial - it is weird to wait for the answers but i guess it is part of the process. I think it is still good to have somebody guiding us specially in the prompt where sometimes it took us 3-4 prompts to get a document we were directly asking for. - Yeah it was quite good experience. I am very happy to get my all questions answered so well.” - I think it is definitely going into the right direction, but it needs a bit more of training. I felt a bit stressed during the audit because I could not predict what we will be told and how it will work.	- After entering a question, ChatGMP was generating a long reply which we had to read through to get the full experience, but it’s felt unnatural and scripted. So it feel like ChatGMP is removing the real life experience of talking to auditees and getting unscripted answers - There is a lot of potential for ChatGMP and I hope that one day it becomes a tool regularly used by companies for external and internal audits. - Really easy to use and very practical - a bit stressful to rephrase the questions while keeping track of answers and documents received - Communication in written mode is decreasing the extent of real interactions a lot. It wasn’t comfortable for the whole group to read the answers. Also, answers were repeating each other. Chat could not give us documents because we used synonyms that weren’t programmed. - I think, ChatGMP is a super cool idea, though it has some problems while understanding the question. Overall, for me ChatGMP is a good AI for auditing
Any suggestions for how to improve this or other additional comments?	the teacher says that “we can’t give you these files because these are stored in the LIMS system”. It will be better if the class can add more parts about this LIMS system, because it is more popular in the industry now, and it can be very helpful if we can know more about it and how to audit LIMS system.	- Maybe it could be Nice if we could talk with it - Make the interface more inviting/ pleasant. It’s a bit basic. - I think, it was bit slow while answering the question. As it is taking time. - it is way slower to write than to talk, so maybe include a copy paste function of the questions. - I think for now it is quite perfect - More time to answer the question. It was very stressful. Make sure that people know we should ask short and precise questions.	- Improve the layout (or interface). Prepare the students for how to prompt. - Add voice recognition for asking questions and audio pronunciations of answers, as it takes too much time to type opening meeting pitch and questions. Add more synonyms so the same questions can be recognized when they are asked in different ways (especially for questions with document requirements). - It just feels that more data has to be analyzed to make chat understand questions and more documents created to provide them. Would be also nice to add any data related to sustainability. It is not part of ICH Q7, but chat can say smth like “its not part of ICH Q7, but we are following next steps that can relate to ICH Q7”

Regarding the quality of the answers both the groups of students performing the audit with the teachers and with *ChatGMP* felt that some answers *made no sense and were not always satisfactory to the auditors*. Therefore *ChatGMP*, although not technically perfect, did not worsen the experience of the students, since the students in the control group underwent the same issues. On the other hand, both groups had positive opinions as well. The control group highlighted the interaction with the teachers and that it was a good real experience overall, while the other group stated that they were impressed and the answers were better than expected, they were shown all the needed documents, that the chatbot was very helpful and that the answers felt like real and informal, which they appreciated.

Multiple students in the control group recognized the potential of *ChatGMP*, which would allow more students to register for the course every year and to reduce the group sizes. They also highlight the limitation of the interaction that would be lacking if the audit was performed with the chatbot, but then reflect that “*which form will be the future eventually depends on the audit objective. The main goal could be just to simulate asking questions and receiving the answers, which the ChatGMP could fulfill, however, the direct interaction with a person during an audit can never be simulated with this audit form. So depending on the accuracy of the audit simulation, either form might be preferred*”. The other groups also agree that there are advantages in either auditee, but that having the possibility of performing the audit exercise with *ChatGMP* would substantially reduce the teachers’ workload. Moreover, they feel that it would be the

same doing the audit with the chatbot or with a teacher. Additionally, two students state that:

- “*I very much enjoyed the whole experience of being the auditor with ChatGMP and working on the report after the audit. This activity familiarized me to many different concepts within GMP. I believe that using ChatGMP will be very helpful to the lectures to perform these audit exercises.*”
- “*Definitely would recommend it, it is the only course with the presence of AI for group exercise, fascinating and innovative. It helps to structure the train of thought. I think it is the future as at a certain point more groups will be able to conduct audits at the same time, and groups will get the same data.*”

Students interacting with *ChatGMP* generally “*think ChatGMP is a super cool idea, though it has some problems while understanding the question. Overall, [...] ChatGMP is a good AI for auditing*”. They indicate some features that should be improved, such as response time, general interface, which is functional but not flawless yet, and the fact that the questions had to be very precise and specific for the chatbot to answer optimally. They suggest to incorporate the option to either copy and paste questions (which is possible but was not allowed by the teacher during the audit) or to utter the question instead of typing it. Another recurring issue found was that some answers were considered not relevant or wrong or that the chatbot sometimes provided the same document multiple times

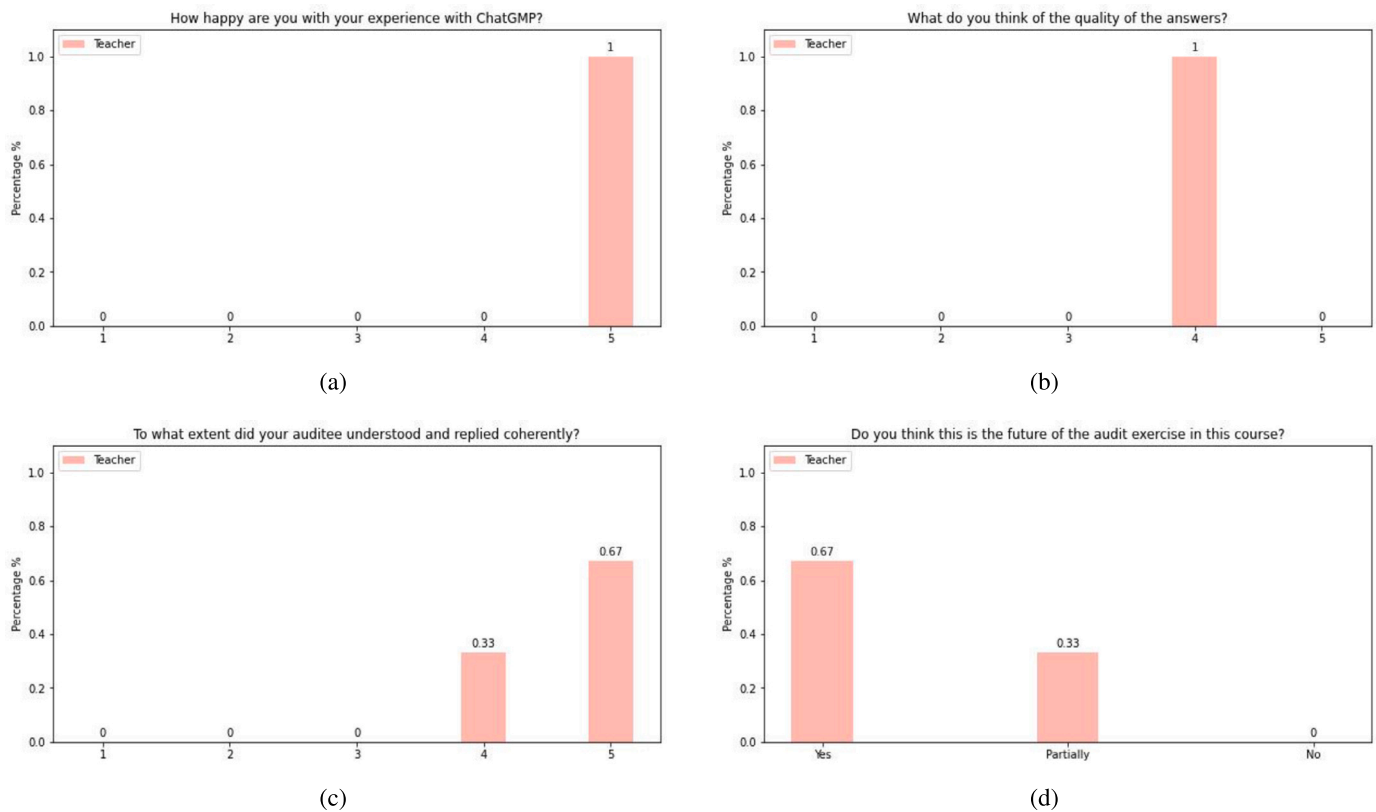


Fig. 6. Responses from the experts of the course (3 teachers). (a) Responses to the question “How happy are you with your experience?”. (b) Responses to the question “What do you think of the quality of the answers generated by the virtual tutor?”. (c) Responses to the question “To what extent do you think that ChatGMP was able to understand your questions and reply coherently?”. (d) Responses to the question “Do you think this is the future of the audit exercise in this course?”. RQ1, RQ2 and RQ3 are on a scale in range 1-5, from lowest to highest.

during the audit. This last comment is believed not to be necessarily negative or something to improve, since questions can be formulated in an multitude of ways and in this context it is preferable for a document to be showed multiple times rather than not enough. Many students state to be impressed with the experience and that the quality of the responses exceeded their expectations. A student describes ChatGMP as “really easy to use and very practical”.

To summarize, there are no significant differences between the students performing the audit with the teachers and ChatGMP, both groups managed to get all the information and documents requested and they had positive feedback overall, as well as constructive suggestions for improvements. This suggests that *ChatGMP* enabled the students to get access to the same information and the same number of documents as the others and to achieve the learning objectives. The students that interacted with *ChatGMP* claim to be satisfied with the initiative and would likely recommend future students to perform the audit with the digital tool.

4.2. Audit results: teachers’ perspectives

To collect an experts’ opinion, as well as to understand whether teachers would approve the experience provided to the students through the interaction with ChatGMP, they were asked to perform a 45 minutes long audit exercise themselves. In particular, they were asked to choose the audit questions from one of the groups they worked with themselves (in the role of an auditee). In this manner, it was then possible to compare the answers previously given by the teachers (when they were the auditee) with the answers of *ChatGMP* (as the auditee). This experiment was run after the teachers had finished all their audit exercises. The teachers’ opinions after interacting with the chatbot are

Table 6

Experts’ review of the response quality of the answers given by ChatGMP.

Corrections	T1	T2	T3	Avg.
No corr. (%)	53.9	62.2	53.1	56.4
Minor corr. (%)	34.6	27.0	12.2	24.6
Major corr. (%)	11.5	10.8	34.7	19.0

reported in Fig. 6. In Fig. 6 (a), all three teachers are very happy about the experience. They also think that the quality of the questions is good, as shown in Fig. 6 (b) and that *ChatGMP* understood and replied very coherently, in 6 (c). Moreover, two out of three teachers believe that *ChatGMP* is the future of the audit exercise, while one believes that it is just partially, as seen in 6 (d).

Table 6 provides a quantitative evaluation of the answers given by *ChatGMP* and reviewed by the experts. Teachers provided feedback on the given answers and were asked to annotate whether they would make no changes, or if they had minor or major corrections. The results show that the majority of the answers given need no corrections, while a smaller set of answers need minor or major corrections. This provides additional evidence that the chatbot, with small improvements, could be a valuable addition to the course and support the role of the teachers, as well as the fact that the quality of the responses given is, on average, sufficiently high. Additionally, Table 7 presents further details on the teachers’ opinions on *ChatGMP*. They all agree that the chatbot could answer well most of the questions (approx. 80%), but there are topics or more complex questions that should have a better answer. Regarding the future of the audit, the teachers approach this question from a more pedagogical perspective, reflecting (rightly so) on “what it means

Table 7

Further elaboration from the experts of the course (3 teachers).

Question	Teacher 1	Teacher 2	Teacher 3
What do you think of the quality of the answers generated by the virtual tutor?	Most of standard questions were well answered (~80%), special questions sometimes are a little off but sometimes even that once are answered in a good manner	Most were spot on or at least very close. The ones not or where I believe improvements could be made.	For sure they have been good. But sometimes it was difficult/impossible to receive a document (management review no. 3). Sometimes they were also a bit “fluffy”
Do you think this is the future of the audit exercise in this course?	See answer below	[...] Regarding the future of the audits I believe this concept is positive and feasible. If the ambition is to have a audit where the teacher is not present, we need to agree and define what this means and how it can be done. There will be a issue about maintenance and possible further development of the system at some time.	It is a good start but I think we have to develop it towards a tool which can understand the questions I am asking via audio not keyboard. I would also prefer a video based answering.
What do you think of the digital audit overall? Where you impressed/ disappointed? Could you please elaborate further on your experience, including any thoughts and reflections?	I think ChatGMP is able to give the students good enough material to work on their audit report. We have to think on how we can keep the situation of a real audit (sitting with them in a room and having time pressure). If they do it from home only online maybe the experience is different and not so “real”	Very impressed. I was particularly impressed that in many cases where a question was complex or maybe not exact, the answer was relevant and coherent with our documents.	Impressed. Very much. I did not expect such effective answering in the first edition...
Is there anything that we should improve?	Sometimes it was not given the document and I needed to ask it specifically. I as a teacher know what documents are available and can ask for them specifically student don't know so there is a chance they are not getting it although they ask a question related to it.	There may still be some specific (rare) questions inspired by ICH Q7 which need a relevant answer. I am committed to try to do a list of these. As mentioned if the audits have to be done without a teacher present, we need to consider how. (In this situation we will obviously not be able to evaluate the performance of the team during the audit.) And eventually the instructions for the teams of how to phrase questions to the ChatGMP will need to be elaborated.	The documents should be easier to get. And I think we should continue training the algorithm. Then I suppose it will be more effective the next time...

to do the audit and how it can be done” and expressing the need to establish whether it should be online or a teacher should still be present. Among the limitations, similarly to the students, they experienced that some documents were more tedious to be shown and suggested to extend ChatGMP's knowledge on certain topics. Moreover, they agree that ChatGMP gives “good enough material to work on their audit report” and that they are “very impressed. [...] particularly impressed that in many cases where a question was complex or maybe not exact, the answer was relevant and coherent with our documents”, and that they “did not expect such effective answering in the first edition”. Finally, the main course responsible stated that:

- “I have evaluated the TW4 [team work 4] reports from the 3 teams [z, y and z] who did the audit via ChatGMP and I have found no difference in overall quality of the audit reports compared to the 6 other TW4 reports I have evaluated so far based on the physical audits with me as auditee. [...] Team z have done the best audit report amongst the 9 I have evaluated so far [...]. The two other ChatGMP reports, one was good average and the other a little below average, but absolutely nothing that differs from the other 6 reports. In general, I can see the ChatGMP teams all get the relevant information according to their audit questions, and the quality of the report is mainly based on the teams' effort and competences to make a quality report based on the available information. My conclusion, also after trying the ChatGMP bot myself on a teams' question list, is that ChatGMP with very minor improvements has sufficient quality and complexity to act as an auditee for our exercise. The minor improvements are related to the fact that ChatGMP it is not 100% accurate in all responses, and in a few cases directly misleading, but this can be solved”.

5. Discussion

The impact of introducing AI in education was thoroughly investigated in many studies, including Yan et al. (2024) and Caccavale et al. (2024a). This section aims to reflect on the experience presented. It is divided in five parts, where the following is discussed: (i) the key learnings, including ChatGMP's strengths and weaknesses, (ii) the new skill-set required for students to acquire, (iii) some suggestions for teachers willing to embed AI in their courses, (iv) possible ethical concerns, and (v) final reflections on the experiment.

5.1. Key learnings and takeaways

Both students and teachers are generally satisfied with their experience with ChatGMP. Students highlight the novelty aspect of the work, describing it as “fascinating and innovative” since this is the first course in their curriculum that uses AI for group exercises. Others believe that “there is a lot of potential for ChatGMP” and hope that “one day it becomes a tool regularly used by companies for external and internal audits”. Overall, the results show no significant difference in the information provided by the teachers and the model. This suggests that ChatGMP enabled the students to get access to the same number of documents as the others and to achieve the learning objectives. The students that interacted with ChatGMP claim to be satisfied with the initiative and would likely recommend future students to perform the audit with the digital tool. However, both students and teachers observed various strengths and weaknesses of the model, which are further discussed in subsections 5.1.1 and 5.1.2.

5.1.1. Strengths of ChatGMP

In the survey, multiple students admit to be impressed by the chatbot, and that the answers were better than they firstly anticipated. They acknowledge that it would reduce the teachers workload and it could be an efficient tool to enable all groups to perform the audit at the same time (if deployed online). Some express that their support towards using *ChatGMP* is motivated by the fact that, by introducing the chatbot, the number of group members could be reduced. This would facilitate the group work and allow students to take more responsibilities and be further involved in the project, perhaps ultimately learning more. They also reflect on the fact that the chatbot would allow for more students to enroll in the course, since, as previously mentioned, 25% more students are rejected every year due to the limited number of teachers available for the audit exercise. They also discuss the fact that *ChatGMP* is easy to use and practical, and appreciate the fact that it replied “as a real person”. Students overall believe that it was a good experience and seem to agree that “it is definitely going into the right direction”.

5.1.2. Weaknesses of ChatGMP

Among the current flaws of *ChatGMP*, the response time and the fact that students were not allowed to paste or utter their questions are the most frequently mentioned. Students also state that at times the responses are not relevant, or directly wrong in some cases. Moreover, *ChatGMP* sometimes generates the same answers for different questions or shows the documents on different occasions. Some groups observe problems with specific documents, and the question had to be rephrased multiple times for a document to be shown. Some students stated that it felt unnatural to perform the audit with the chatbot and that the interaction with the teacher is an important aspect of this exercise. In the survey, it is also highlighted that some students found asking questions to be stressful, since they had to be quite short and precise. Finally, it was mentioned that the interface could be improved.

We are aware of the current limitations of *ChatGMP*, which are further discussed in Section 7. However, given that this was the first test run, it can be considered a successful experiment. All the collected feedback will be used to improve the model before using it again in the course in spring 2025.

5.2. New skills required

As suggested in Table 5, students learned that being able to effectively prompt is a necessary skill for ensuring a successful interaction with a chatbot and that it could lead to substantially better responses. In fact, they discovered that the elaborate and long questions that they prepared before the audit exercise were not appropriate for *ChatGMP* and that shorter, simpler, and clearer questions worked better and more efficiently. Moreover, we argue that if a course aims to include an AI tool such as the one proposed in this work, a very general introduction to AI and generative chatbots in particular, should be provided to allow students to better understand how answers are generated and how their data is treated. Being able to understand how an AI model is trained and used would also spark their critical thinking, enabling them to reflect on whether it is being applied in a correct (and hopefully ethical) manner or whether further adjustments should be conducted. In summary, under this new paradigm, in order to obtain the best results when interacting with LLMs provided as chatbots, and in the most efficient and ethical way, the users/students must: (i) learn how to query (aka prompt engineering) - being able to clearly articulate questions is paramount for successful use of chatbots; (ii) develop critical thinking and problem-solving capabilities; (iii) have basic AI literacy - understanding the basic AI concepts can aid in making sense of information provided by generative AI; and, (iv) be aware of bias and privacy concerns - being responsible and ethical when using generative AI.

5.3. Directions for teachers and researchers

Given the previously discussed new skill-set required by the students to face the “AI Revolution”, teachers need to keep up with the times and contribute to including these skills in the curriculum. For example, as shown by the students’ responses in Table 5, learning how to effectively craft the prompt given to an LLM is fundamental for the success of this as well as other applications. So it is important for teachers wishing to include generative chatbots to their courses to carefully teach students to prompt, e.g., by providing examples in class. Additionally, teachers need to update the learning objectives to incorporate possible changes introduced by an AI application and define what is considered to be the “success” criteria.

As pointed out in Table 7, another consideration worth mentioning is that, if a chatbot replaces a normal interaction between teacher-student, it is necessary to establish how the experience can be practically administered. Possible solutions could be to either deploy the platform online, so that students would be able to access it at any time, or to deploy it locally, meaning that a responsible would be needed to facilitate the interaction.

As previously described, *ChatGMP* leverages a pre-trained, already developed LLM, FLAN T5 (Chung et al., 2022). This model, and dozens of similar ones, can be easily downloaded from libraries such as the HuggingFace. This certainly democratizes the use of AI (for the average researcher) to some extent. In fact, until a few years ago, researchers arguably were required to possess a much deeper technical knowledge of AI architectures and their building blocks to “make their models work”. Currently, it is possible to almost instantly load a model in Python and use it. However, for custom applications such as the one presented in this study, programming skills, as well as technical knowledge about AI models, are essential. In this project, the data curation process was the most technically difficult part, since many steps had to be performed to ensure properly pre-processed data. Therefore, it could perhaps be generalized that if data has to be collected, the difficulty level of the task increases significantly. On the other hand, if the data is available and in a ready-to-use format, the process of creating a custom chatbot can be accelerated substantially.

This study presented a model fit to answer questions related to a fictitious company about their GMPs. However, *ChatGMP* would be able to similarly answer less case-specific questions if real GMP data was available, without having to be re-trained. This is due to the semantic search algorithm, which is able to find the context in the given data, regardless of whether it is fictitious or real. This suggests that similar models could undoubtedly be used in various courses and applications to automate repetitive tasks and question-answering processes. Overall, this inherent flexibility and grasp of data that most LLMs present highlights the potential of generative AI models to be applied to a wide variety of fields and applications, including, but not limited to, education.

Finally, the infrastructure for deployment needs to be considered carefully, according to the size of the application and the speed required. These tools need periodic maintenance and upgrades, which should also be taken into consideration. In this study, the infrastructure necessary to run the model is described, and the code is available on GitHub, see Section 9 for further details.

5.4. Ethical considerations

Needless to say, possible ethical concerns regarding the introduction of AI-based tools should be assessed before using these models in an educational context. The focus on ethics and fairness in AI has gained traction in recent years, both by researchers (e.g., Holmes et al. (2021); Slade and Prinsloo (2013); Cerratto Pargman and McGrath (2021); Yang et al. (2021)) and governmental institutions such as UNESCO and the European Parliament. AI models should be used responsibly and should align with fundamental human values and principles to safeguard human well-being (Dignum, 2019). This responsibility should cover all

steps of the AI lifecycle, including data acquisition, model implementation and deployment of systems, as well as how end-users understand and adopt the system. Therefore, before deploying AI systems in education, various aspects of equity, responsibility, fairness, autonomy, accountability, and transparency must be considered, as well as possible concerns that the system might incite (Prinsloo, 2020; Slade & Prinsloo, 2013; Cerratto Pargman & McGrath, 2021; Velander et al., 2021).

5.5. Overall reflections on the experiment

ChatGMP was created with a specific purpose: replacing the auditee, embodied by a teacher, in an audit exercise performed in a Master's Degree course in GMP. This initiative was conducted to enable substantial time-saving, automate a repetitive task and provide an interactive and fun experience for the students. To perform the audit exercise, 21 groups of 5-6 students were formed, and out of these 3 volunteered to perform it with *ChatGMP*. The initial target goal for the number of groups to test *ChatGMP* was set to 3-5 groups, which would ensure a fair evaluation of the platform. No additional groups than the ones that genuinely volunteered were compelled to conduct the audit exercise with the chatbot, since the research group deemed it not to be ethical to force any group towards a certain auditee. Therefore, the fact that 3 groups volunteered can be considered satisfactory and highlights the students' openness and willingness towards this initiative.

Although the platform is not fully automated, and somebody supervising the interaction was still necessary during this edition of the course, *ChatGMP* was able to fulfill its objective: reply to students' questions, provide meaningful information and documents that would enable them to write a complete audit report and reduce the number of audits that each teacher had to conduct. So overall, the experiment can be considered a success. This can be concluded not only from the results obtained by the groups of students in the audit report, which show no statistically significant difference to the ones that performed the audit exercise with the teachers (and on the contrary, the best report was of a group interacting with *ChatGMP*), but also because it was well received by both the students and the teachers. Moreover, all the teachers expressed that they were impressed with the ability of the chatbot to respond to the questions and that the quality of the answers was very high, corroborating the fact that the chatbot can be considered a valuable tool for the course and can become an integral part of the next editions as well.

6. Future perspectives

To investigate what is the best strategy to adopt in this use case, various approaches, such as comparing prompt engineering and instruction fine-tuning, are planned to be benchmarked in the future. Fine-tuning could be beneficial in fields that use language in an idiosyncratic manner, such as in our specific case.

Future research could benefit from incorporating more objective evaluation metrics to assess the quality of the generation and a larger sample size. For instance, future studies could measure student performance on the reports or use different metrics to automatically evaluate the quality and relevance of the answers provided.

Moreover, in the future further capabilities will be added to the platform. For example, we plan to add a reaction button to the responses provided by the chatbot, so that students could evaluate right away whether the answer is helpful and relevant or the chatbot is lacking knowledge. Another idea is to allow users to speak to the chatbot instead of typing, and for the chatbot to utter a response back. This would result in adding a speech-to-text (for the user's questions) and text-to-speech (for the chatbot) step to the pipeline. Finally, the final version of the model is planned to be a human-like agent or an avatar, so that the experience could feel more "real" to the students.

7. Limitations

This study has several potential limitations. Firstly, the analysis is based on a small sample. Secondly, the findings may have restricted generalizability due to the specific context of the subject taught. Institutional structures and technological landscapes unique to the university this study was applied to may limit how these results can extend to other educational environments. Ongoing developments in generative chatbots and shifts in educational policies could impact the applicability and validity of the results. We also acknowledge that the model used in this work is not the latest released LLM available, and therefore it has its architectural limitations. Many models released in 2024 are able to produce higher-quality generations and are more powerful. However, we were limited by available resources for the project, both in terms of budget and GPUs, so this model was considered to be a good trade-off between fast runtime and still ensuring reliable and safe responses for the students. Lastly, the analysis presented relied on subjective judgments, which can introduce variability in the results due to personal opinions or preferences, as well as individual biases and expertise. This inherent subjectivity highlights the wide variety of perspectives, unwinding the diverse viewpoints and challenging researchers to reflect on their possible consequences.

8. Conclusion

In this work, *ChatGMP*, a chatbot based on the pre-trained FLAN-T5 base model, was introduced. *ChatGMP* is introduced in a Master's Degree course in GMP and is able to perform an audit exercise, answering students' questions and responding with meaningful information and documents. The chatbot was created to enable substantial time-saving for the teachers, automate a repetitive task and provide an interactive and fun experience for the students. For the most part, the experiment was considered a success. This can be concluded not only from the results obtained in the audit report, where the best report was presented by a group performing the audit with *ChatGMP*, but also because it was well received by both students and teachers. On the other hand, the introduction of *ChatGMP* into the course also presented some weaknesses. Some groups observed problems with acquiring specific documents, and the question had to be rephrased multiple times for a document to be shown. Some students stated that the interaction with the teacher is an important aspect of this exercise and therefore it felt unnatural to perform the audit with the chatbot. In addition, some students found asking questions stressful, as the questions had to be fairly short and precise. Finally, it was mentioned that the interface could be improved. Among the limitations of the experiment is that the analysis relied on subjective judgments, which can introduce variability in the results due to personal opinions or preferences, as well as individual biases and expertise. For example, the teachers' perspective may be slightly biased by the fact that introducing *ChatGMP* could allow more students to enroll in the course, or it could mean that they would have to perform fewer audit exercises.

Overall, all the teachers expressed that they were impressed by the chatbot and the quality of the responses generated, corroborating the fact that *ChatGMP* can be considered a valuable tool for the course and can become integral part of the next editions.

9. Code availability

This study is committed to the open source way set of principles. The code, together with the pre-processed data, including the audits in QA format and QA pairs extracted from the supporting documents such as the available documents of "Pharma A/S", is freely available at <https://github.com/FiammettaC/ChatGMP>. The code is published under the MIT license. Currently, it is compatible with Python 3.x, for Windows and Linux operating systems. It is also recommended to use a conda environment to run the code.

CRediT authorship contribution statement

Fiammetta Caccavale: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Carina L. Gargalo:** Writing – review & editing, Supervision, Resources, Project administration. **Julian Kager:** Writing – review & editing, Validation, Supervision, Methodology. **Steen Larsen:** Writing – review & editing, Validation, Supervision, Project administration. **Krist V. Gernaey:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition. **Ulrich Krühne:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Statements on ethics and open data

Informed consent was obtained from all participants, and their privacy rights were strictly observed. Prior to their participation, all participants were informed that the collected data would be used for research purposes and would be handled anonymously. Detailed information about the study was provided to participants before beginning the survey, reassuring them of their right to voluntary participation and withdrawal. The data can be obtained by sending an e-mail request to the corresponding author.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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